* Assignment 2 *

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Course : TOC

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Q1 Write closure properties of regular languages.

A. Complement :

If L is regular, its complement is also regular.

[
formline L = \sigma^* - L
]

2. Kleene star :

If L is regular, then (L^*) is also regular.

Example

If $(L = \{ab\})$, then $L^* = \{\text{epsilon}, ab, abab, \text{etc}\}$.

3. Reversal :

If L is regular, its reverse (L^R) is also regular.

example : if $(L = \{a, b\})$, then $(L^R = \{ba\})$

4. Substitution :

Replacing each symbol of regular language with a regular language produces another regular language.

Q2 Construct an NFA with ε-transitions for the regular expression (a+b)*(ab).

Given:

[

R = (a+b)*(ab)*

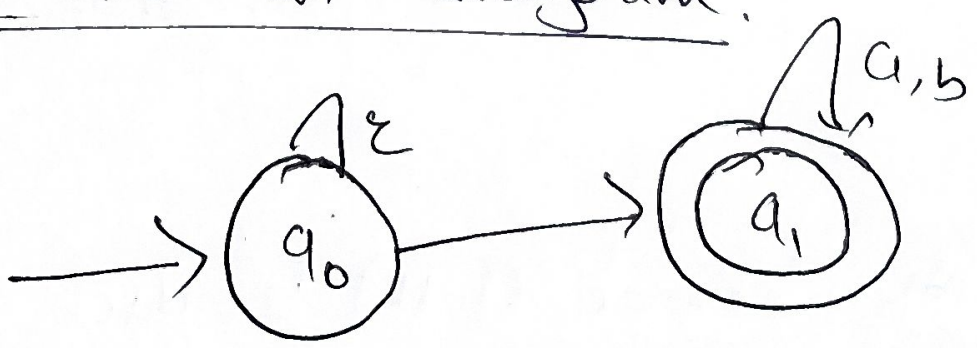
]

Since ((a+b)*) accepts every string over {a,b}, the complete expression also represents all strings over {a,b}.

NFA States:

- Initial State: q_0
- Final state : q_1
- Alphabet: (Sigma = {a,b})

Transition diagram.



Transition diagram:

State	a	b	ε
→ q_0	∅	∅	(q_1)
→ q_1	(q_1)	(q_1)	∅

Result :- The ε-NFA accepts all strings over {a

Q3 Give an informal description of the language denoted by $(0+1)^*00$.

Given:

$$L = \{R = (0+1)1^*00\}$$

Description:

- $(0+1)$ represents either 0 or 1.
- $(0+1)^*$ represents any combination of 0 & 1.
- 00 represents the last two symbol

Accepted Strings:

00, 000, 100, 1100, 0100.

Rejected String:

01, 10, 11, 101.

Conclusion:

The language contains all binary strings ending with two consecutive zeros.

$$L = \{w00 \mid w \in (0,1)^*\}$$

Q4 Prove that the language having equal numbers of a's & b's is not regular using pumping lemma.

Given:

[
L = {w | w in (a+b)* n_a(w) = n_b(w)}
]

Step 1: Assume L is regular.

Step 2: Let p be the pumping length.

Choose:

[
w = a^p b^p
]

Step 3: Divide the strings.

[
w = xyz
]

According to pumping lemma.

1. (|xy| <= p)
2. (|y| > 0)
3. (xy^i z in L)

Q5 Apply Arden's Theorem & drive the regular expression For the given automata

Soln

$q_1 = R_1 = \epsilon + R_1(0)$

$q_2 = R_2(1) + Q_2(1)$

$q_3 = R_3 = R_2(0) + R_3(011)$

Applying Arden's theorem.

$\boxed{R_1 = 0^*}$

For R_2:

$R_2 = R_1(1) + R_2(1)$

$R_2 = 0^*1 + R_2(1)$

$\therefore \boxed{R_2 = 0^*11^*}$

Final Stage

$R = R_1 + R_2$

$= 0^* + 0^*11^*$

$\boxed{R = 0^*1^*}$

Since ($|xy^2|$ leq p), y contains only a's.

let ;

$$x = a^k, \quad k \geq 0$$

Step 4: Take (i=0).

$$xy^0z = a^{p-k}b^p$$

Number of a's = p-k.

Number of b's = p

Both are not equal.

Hence, ($xy^0z \notin L$).

This contradicts the pumping lemma.

Conclusion:

L is not regular.